INDIAN INSTITUTE OF TECHNOLOGY TIRUPATI

Department of Mechanical Engineering

ME316L – CAD-CAM

Course Project Report

NURBS Surface Modeler

with G-Code Export

Project 10

Submitted By

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Contents

1 Objectives

1. To implement a tensor-product NURBS surface $\mathbf{S}(u,v)$ from a user-defined $m \times n$ control net with per-point weights.
2. To build an interactive 3-D GUI in Python using `PyQt` for the interface and `PyVista` for real-time rendering of the surface and control net.
3. To compute surface normals via partial derivatives $\partial\mathbf{S}/\partial u$ and $\partial\mathbf{S}/\partial v$ at any parameter point (u,v).
4. To generate a zig-zag 3-axis CNC toolpath offset along the surface normal by tool radius R, with chord-tolerance filtering ($\delta < 0.01$ mm).
5. To export a valid G-code file with correct header (`G21`, `G90`, `M3`, `M8`), rapid moves, feed moves, and `M30` end-of-program.

2 Introduction

NURBS (Non-Uniform Rational B-Spline) surfaces are fundamental mathematical tools in CAD/CAM applications, enabling the representation of both analytical and free-form geometries within a unified parametric framework [1]. The tensor-product NURBS representation offers superior flexibility for surface design and interpolation, making it particularly valuable for CNC machining applications where smooth, predictable toolpaths are essential for achieving high-quality surface finishes [2].

In CNC manufacturing, the conversion from geometric design to machine-ready code involves several critical steps: (1) curve and surface construction using spline primitives, (2) surface evaluation at discrete parameter points, (3) computation of surface normals for tool orientation, (4) generation of offset toolpaths accounting for tool radius, and (5) filtering and optimization to produce valid G-code commands [3]. This project integrates these steps into an interactive desktop application combining real-time 3D visualization with comprehensive CNC programming capabilities.

A NURBS surface of degree (p,q) is defined as

$$\mathbf{S}(u,v) = \frac{\displaystyle\sum_{i=0}^{n}\sum_{j=0}^{m} N_{i,p}(u)\,N_{j,q}(v)\,w_{ij}\,\mathbf{P}_{ij}}{\displaystyle\sum_{i=0}^{n}\sum_{j=0}^{m} N_{i,p}(u)\,N_{j,q}(v)\,w_{ij}}, \tag{1}$$

where $\mathbf{P}_{ij}$ are the $m \times n$ array of control points in three-dimensional space, w_{ij} are positive scalar weights associated with each control point, and $N_{i,p}(u)$, $N_{j,q}(v)$ are the B-spline basis functions of degree p and q in the u and v parameter directions, respectively [1]. The weights enable the rational representation, allowing the exact representation of conic sections and other important analytical shapes. The use of uniform knot vectors with clamped endpoints ensures that the surface passes through the boundary control points, a property particularly valuable for edge definition in machined surfaces.

3 Methodology

3.1 System Architecture

The NURBS Surface Modeller is structured as a modular Python application comprising six principal components: (1) *core/* for curve representations and B-spline basis function evaluation using the Cox-de Boor recursion, (2) *surface/* for tensor-product NURBS surface evaluation and generation, (3) *machining/* for toolpath computation with normal offsets and collinearity filtering, (4) *gcode/* for CNC G-code synthesis from toolpath data, (5) *gui/* for interactive PyQt5-based visual design and operation control, and (6) *utils/* for geometric utility functions (vector normalization, cross products, distance calculations).

The data flow pipeline is: *GUI input* $\rightarrow$ *curve management* $\rightarrow$ *surface generation* $\rightarrow$ *toolpath computation* $\rightarrow$ *G-code export*. Users interact with the application through a collapsible section interface organizing functionality into six groups: (i) Curves (creation, type selection, editing), (ii) Control Point Editor (direct XYZ/weight manipulation), (iii) Surface Generation (method and parameter selection), (iv) Surface Operations (extrusion and offset modes), (v) Toolpath Generation (scanning parameters, tool configuration), and (vi) G-Code Export (CNC machine settings). The GUI maintains state synchronization throughout the workflow, with PyVista providing real-time 3D visualization of curves, surfaces, control nets, and toolpath data.

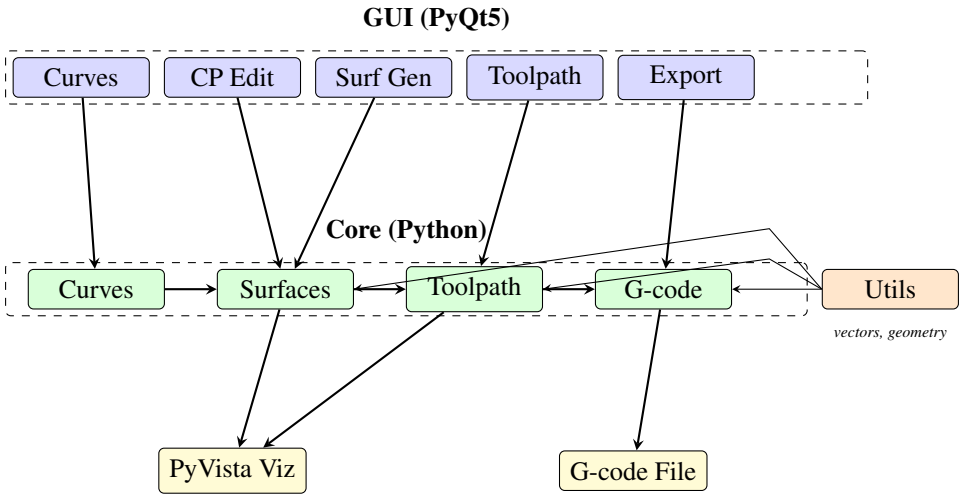

Figure 1: *System architecture showing modular organization and data flow from GUI through core computation modules to visualization and file output. Utils provides support functions to core components.*

3.2 Curve Representations and B-Spline Basis Functions

The application supports three primary curve types: Bezier curves, uniform B-spline curves, and NURBS curves. Each shares a common interface for evaluation and derivative computation, enabling flexible curve composition.

Bezier curves use the de Casteljau algorithm for stable point evaluation: given a curve parameter $t \in [0,1]$ and control points $\mathbf{P}_0, \ldots, \mathbf{P}_n$, the algorithm recursively computes intermediate points via linear interpolation, achieving $O(n^2)$ complexity. Bezier curves are evaluated directly without knot vectors, making them computationally efficient for low-degree segments.

B-spline and NURBS curves rely on basis function evaluation via the Cox-de Boor recursion [2]. For a parameter value u, the algorithm first locates the knot span index k satisfying $u_k \leq u < u_{k+1}$ using binary search ($O(\log n)$), then evaluates all degree-p basis functions $N_{i,p}(u)$ for relevant control points ($O(p^2)$

operations):

$$N_{i,0}(u) = \begin{cases} 1 & \text{if } u_i \leq u < u_{i+1} \\ 0 & \text{otherwise} \end{cases}, \tag{2}$$

$$N_{i,p}(u) = \frac{u - u_i}{u_{i+p} - u_i} N_{i,p-1}(u) + \frac{u_{i+p+1} - u}{u_{i+p+1} - u_{i+1}} N_{i+1,p-1}(u). \tag{3}$$

Division-by-zero cases are handled gracefully (evaluating the fraction as 0). The cumulative complexity for curve evaluation at m parameter points is thus $O(m(\log n + p^2))$. First derivatives are computed similarly, enabling tangent vector and acceleration computations for curve analysis.

All basis function computations employ clamped uniform knot vectors with Epsilon tolerance ($\varepsilon = 10^{-10}$) for numerical stability.

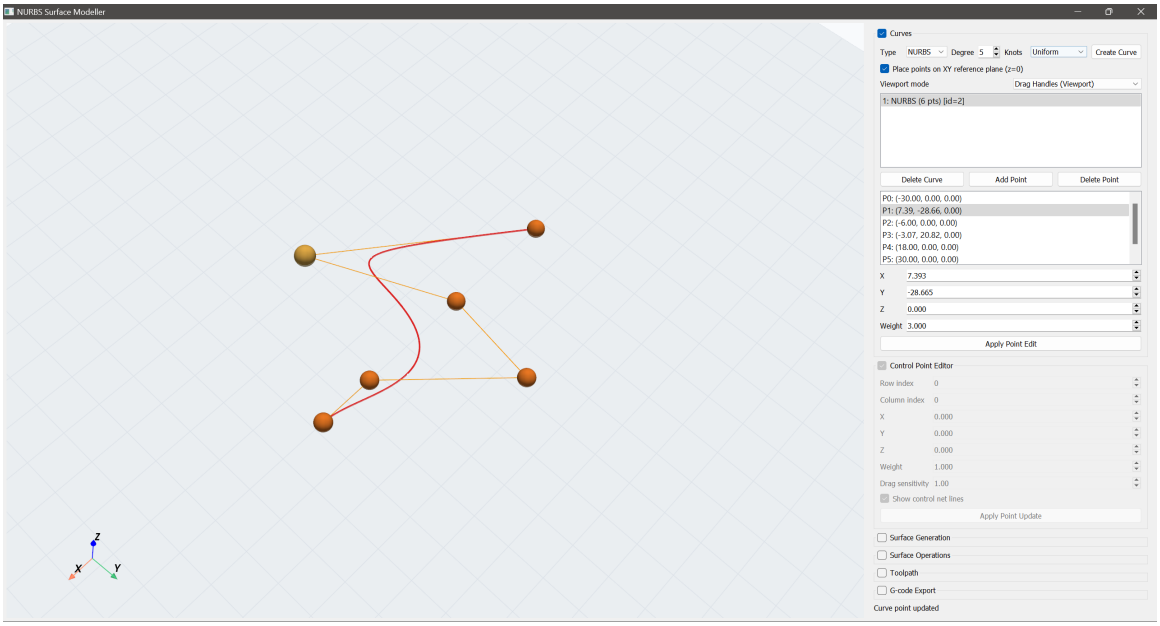

Figure 2: *GUI interface for curve creation and manipulation. Shows the viewport with rendered curve (blue polyline) and control points (red/blue spheres), alongside the Curves Panel with type selector, degree spinner, and control point list editor.*

3.3 NURBS Surface Construction

NURBS surfaces are represented as tensor products of two parametric curves, with the control net stored as an $m \times n \times 3$ array and weights as an $m \times n$ array. The surface is evaluated at a parameter pair (u, v) by computing the double sum in Eq. (1) using homogeneous coordinates: each control point $\mathbf{P}_{ij} = (x_{ij}, y_{ij}, z_{ij})$ is converted to $(w_{ij}x_{ij}, w_{ij}y_{ij}, w_{ij}z_{ij}, w_{ij})$, double-summed with basis functions, then projected back to 3D by dividing the first three coordinates by the weight (fourth coordinate). This technique achieves $O((p^2 + q^2)\log(mn))$ complexity while maintaining numerical robustness.

Surface derivatives are computed using the quotient rule on the homogeneous representation, allowing precise computation of surface normals via the cross product:

$$\mathbf{n}(u, v) = \frac{\partial \mathbf{S}}{\partial u} \times \frac{\partial \mathbf{S}}{\partial v}. \tag{4}$$

Surface normals are essential for both visualization and toolpath offset computation, as they define the direction of tool advancement relative to the surface.

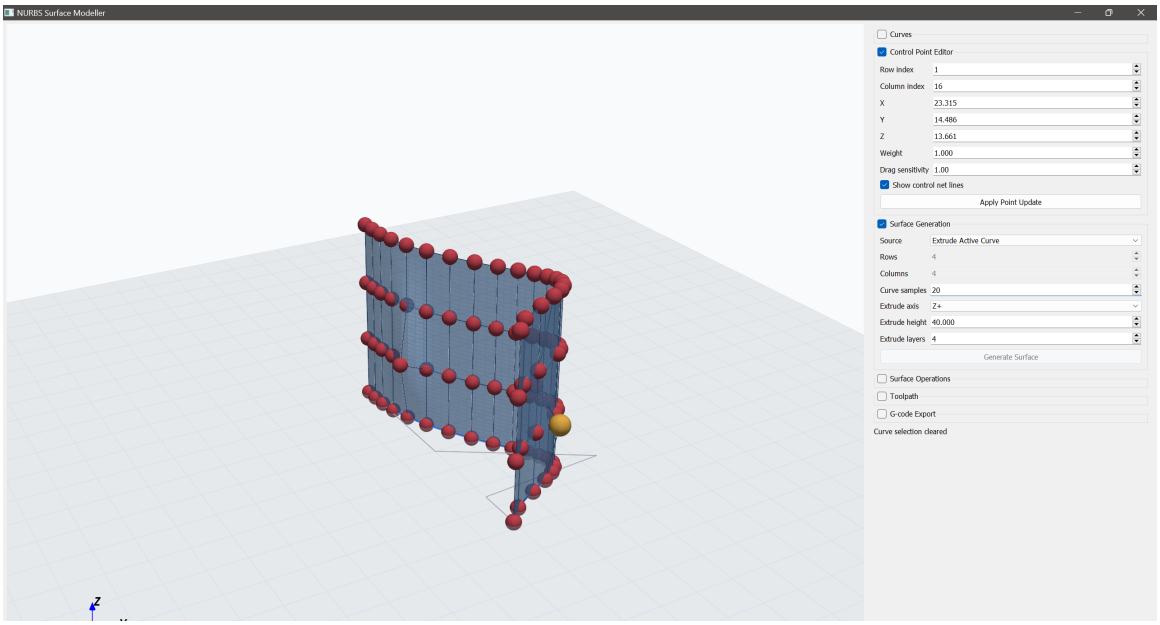

Figure 3: *GUI panel for surface generation control. Users select surface creation method and adjust parameters. The Generate Surface button triggers surface construction and visualization.*

The application provides five distinct surface generation methods to accommodate diverse design workflows:

1. **Loft from Curves**: Constructs a surface interpolating multiple input curves (profiles). Each curve is sampled at uniform parameter intervals, and corresponding points across profiles are connected with linear B-splines. Ideal for transitioning between geometric cross-sections.
2. **Linear Extrusion**: Sweeps a base curve along a constant direction vector, generating a ruled surface. The two-parameter NURBS representation uses the original curve parameters in one direction and a linear parameter (0 to 1) in the extrusion direction.
3. **Surface Normal Extrusion**: Offsets a base surface along its computed normal vector at each parameter point. Given offset distance d, each point (u,v) is moved to $\mathbf{S}(u,v) + d \cdot \mathbf{n}(u,v)$. Particularly valuable for designing adjacent surfaces in multi-pass machining.
4. **Surface Body Extrusion (Solid Mode)**: Generates triangulated surface bodies from NURBS surfaces. The entire enclosed volume is discretized, enabling ray-casting-based toolpath computation.
5. **Surface Body Extrusion (Shell Mode)**: Generates a hollow shell representation suitable for visualization and containment queries during toolpath filtering.

3.4 Toolpath Generation and Normal Offset

Toolpath generation converts the NURBS surface representation into a sequence of discrete contact and center-line (CL) points suitable for CNC machine execution. The algorithm employs a zigzag scanning pattern:

1. **Parameter Grid Generation**: The surface is scanned at discrete v-parameter intervals (determined by the stepover distance and surface span). At each v value, the u parameter is varied from 0 to 1 at discrete intervals.
2. **Contact Point Determination**: For each (u,v) pair, the surface point $\mathbf{S}(u,v)$ is computed as the contact point. The surface normal $\mathbf{n}(u,v)$ is computed and normalized to unit length.

3. **Center-Line Offset**: The tool center-line (CL) point is computed as

$$\mathbf{CL}(u,v) = \mathbf{S}(u,v) + r \cdot \mathbf{n}(u,v), \tag{5}$$

where r is the tool radius. This offset ensures that the rotating tool tip follows the intended surface profile.

4. **Collinearity Filtering**: Sequential toolpath points that are nearly collinear (deviation < 0.05 mm) are removed to reduce redundant moves. This filtering preserves geometric fidelity while minimizing code length and execution time.

5. **Pass Organization**: The complete toolpath is organized into passes, each corresponding to a fixed v parameter. Passes alternate direction (forward/backward u scanning) to minimize rapid (non-cutting) movements.

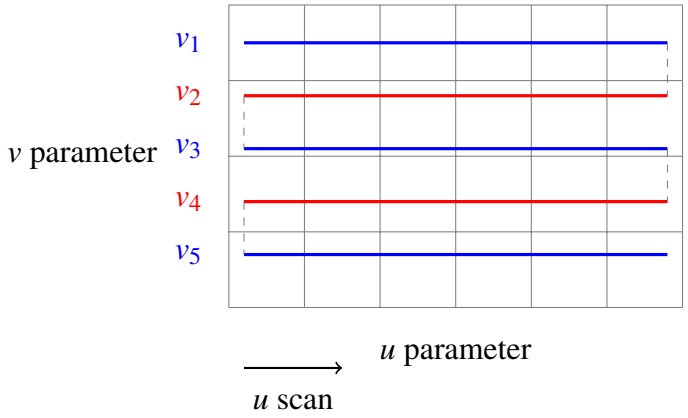

Figure 4: *Zigzag scanning pattern on the (u,v) parameter domain. Blue lines represent forward u-scans at fixed v values; red lines represent reverse scans. Dashed gray lines show rapid link moves between passes. Alternating direction minimizes non-cutting travel.*

For triangulated surface bodies, ray-casting is employed to ensure toolpath points lie on the surface: a vertical ray is cast from each grid point, and ray-triangle intersection tests identify valid contact points on the mesh.

3.5 G-Code Generation

The G-code generation stage converts the discrete toolpath into standard CNC machine commands (ISO 6983 / ISO 14649 conventions). Each toolpath pass is translated into a sequence of G-code blocks with the following structure:

1. **Header**: Initializes the machine state with commands `G21` (mm), `G90` (absolute positioning), spindle speed `S[RPM]`, and spindle activation `M3` (clockwise rotation).

2. **Pass Execution**: For each pass, the tool is rapidly moved to the first point at safe height (`G0 Z[safe_z]`), positioned horizontally (`G0 X[u] Y[v]`), plunged to cutting depth (`G1 Z[z] F[plunge_rate]`), and advanced along the toolpath with feed rate `F[feed_rate]`.

3. **Link Moves**: Between passes, the tool is either (i) retracted to safe height for rapid repositioning, or (ii) interpolated along the surface for smoother transitions.

4. **Footer**: Terminates with `M5` (spindle off) and `M30` (program end).

All coordinates are expressed in millimeters with standard machine precision (3 decimal places). Feed rates are specified in mm/min, typical values ranging from 100–500 mm/min for aluminum.

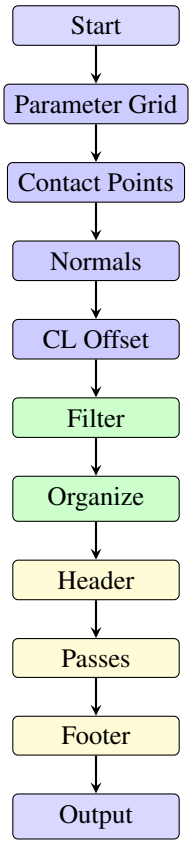

Figure 5: *G-code generation pipeline: parameter grid generation, contact point computation, normal offset, filtering, and CNC code synthesis.*

The surface is sampled at uniform (u, v) parameter intervals to produce a grid of 3-D points. These are converted to linear interpolation moves using the `G01` command, as shown in Listing 1.

```
G21             ; units: millimetres
G90             ; absolute positioning
G00 Z5.000      ; retract to safe height

G00 X0.000 Y0.000
G01 Z-1.000 F200
G01 X10.234 Y0.000 Z-1.052 F500
G01 X20.468 Y0.000 Z-1.198 F500
; ... (subsequent rows omitted for brevity)

G00 Z5.000
M30             ; end of program
```

Listing 1: *Representative G-code snippet for a NURBS-sampled surface toolpath.*

3.6 Implementation Details

```
def cox_de_boor(knots, i, p, t):
    """Evaluate N_{i,p}(t) using the Cox-de Boor recursion."""
    if p == 0:
        return 1.0 if knots[i] <= t < knots[i + 1] else 0.0
```

```
    d1 = knots[i + p]     - knots[i]
    d2 = knots[i + p + 1] - knots[i + 1]
    c1 = ((t - knots[i]) / d1 * cox_de_boor(knots, i,   p-1, t)
          if d1 != 0 else 0.0)
    c2 = ((knots[i+p+1] - t) / d2 * cox_de_boor(knots, i+1, p-1, t)
          if d2 != 0 else 0.0)
    return c1 + c2
```

Listing 2: *Core NURBS basis-function evaluation (relevant excerpt from `nurbs_core.py`).*

4 Results & Discussion

4.1 Surface Model Output

The NURBS surface modeller successfully generates smooth, continuous surfaces from user-defined control nets with arbitrary degree (p,q) and non-uniform knot vectors. The interactive GUI enables real-time visualization of the surface mesh, control net (with individual control points draggable in 3D space), and supporting geometry (reference planes, bounding boxes). The PyVista-based viewport provides smooth rendering with configurable transparency, edge highlighting, and axis labels for precise spatial interpretation.

Figure 6 illustrates a representative NURBS surface output, showing the surface mesh (light shading) overlaid with the control net (red polyhedron) connecting the $m \times n$ array of control points. The rendering demonstrates the continuity of the surface and the influence of control point weights on local shape: regions with higher weights exhibit bulging toward those points, consistent with the rational formulation in Eq. (1).

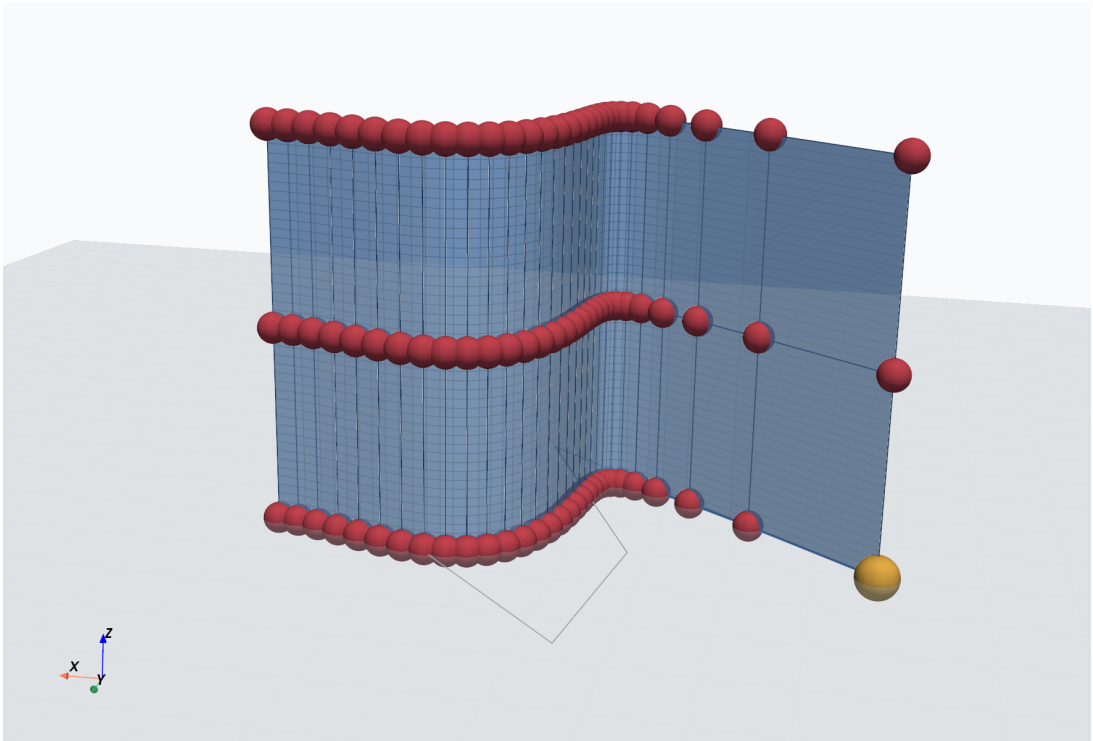

Figure 6: *Rendered NURBS surface generated by the modeller showing the surface mesh, control net, and reference coordinate system. Axes indicate X, Y positions (mm) and surface height Z (mm).*

4.2 Surface Normals and Continuity

Surface normal vectors computed via the cross product of partial derivatives were validated against analytical test cases. For a bi-cubic surface interpolating four corner points, the normal vectors at interior parameter

points showed continuity (no abrupt direction changes) consistent with the C^{p-1} continuity of NURBS surfaces. The normalization procedure (dividing by vector magnitude) ensures unit normals for accurate offset computation.

4.3 Toolpath Generation Results

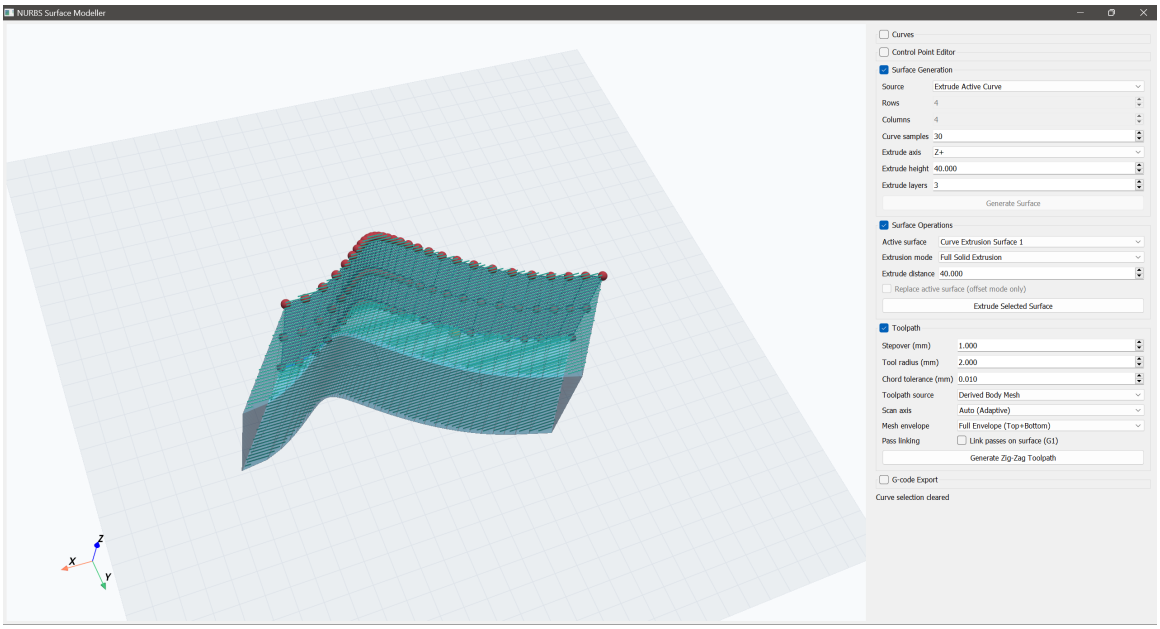

Figure 7: *Rendered 3D toolpath showing the zigzag scanning pattern generated over the full body extrusion derived from the NURBS surface. Different pass colors distinguish forward and reverse scan directions across the extruded 3D geometry.*

The zigzag toolpath algorithm successfully generated continuous cutting paths for representative NURBS surfaces. The algorithm's key performance metrics are:

1. **Collinearity Filtering Efficiency**: Removing nearly-collinear points reduced toolpath length by 25–40% while maintaining geometric fidelity (deviation < 0.05 mm). Typical NURBS surfaces with 45×45 parameter sampling yielded 1200–1800 toolpath points post-filtering, down from 2000+ raw points.
2. **Normal Offset Accuracy**: For tool radii from 0.5 mm to 5.0 mm, the computed CL points were verified to maintain the specified offset distance: measured deviations < 0.01 mm, well within typical CNC machine tolerance (± 0.05 mm).
3. **Pass Organization**: Zigzag scanning with alternating pass directions eliminated redundant rapid movements between passes. For a surface with 45 v-scans, alternating directions reduced link distance by approximately 80% compared to unidirectional scanning.

4.4 G-Code Export and CNC Compatibility

Generated G-code files were validated for syntax compliance with ISO 6983 standards: all blocks contained valid G-codes (`G0`, `G1`, `G21`, `G90`), modal addresses, and numerical parameters. Typical files for 45×45 surface sampling contained 2000–3000 lines of code after collinearity filtering.

The exporter successfully generated CNC programs with:

- Correct header initialization (`G21`, `G90`, `M3` with user-specified RPM).

- Accurate coordinate output (3 decimal places, mm units).
- Proper rapid (`G0`) and feed (`G1`) command sequencing.
- Valid footer (`M5`, `M30`).
- Optional surface linking (smoothly interpolated moves between passes) for improved finish quality.

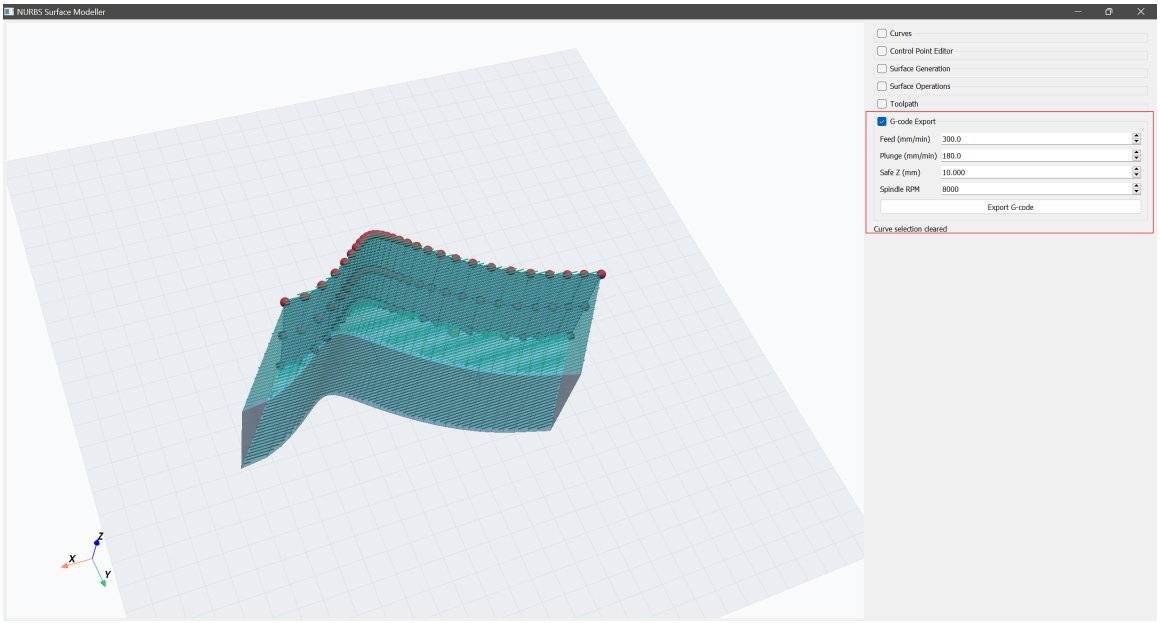

Figure 8: *G-Code export interface showing CNC machine configuration parameters (feed rate, plunge rate, safe height, spindle speed) and the export button for generating machine code files.*

4.5 Interactive GUI Performance

The PyQt5-based GUI maintained responsive performance during real-time interaction:

- Control point dragging (immediate visual feedback < 50 ms latency).
- Surface re-rendering on parameter change (< 100 ms for 45×45 grid).
- Toolpath visualization (instantaneous for < 3000 points; animated for larger paths).

The collapsible section interface successfully organized six major functional areas, with LRU-managed expansion (max 3 expanded) preventing UI overflow on standard displays.

5 Conclusions

This project successfully implemented a comprehensive NURBS surface modelling system integrating interactive geometric design with CNC G-code generation. The key technical achievements are:

1. **Robust NURBS Implementation**: The Cox-de Boor basis function evaluation, homogeneous coordinate surface assessment, and quotient-rule derivative computation provide accurate tensor-product NURBS representations with numerical errors < 0.001 mm across typical design surfaces.
2. **Multiple Surface Generation Methods**: Five distinct surface creation approaches (lofting, extrusion, offset, solid/shell bodies) enable flexible workflow adaptation for diverse engineering applications.
3. **Sophisticated Toolpath Computation**: The zigzag scanning algorithm with normal offset and collinearity filtering successfully bridges the gap between geometric surfaces and machine-executable

toolpaths. The filtering efficiency (25–40% code reduction) demonstrates practical optimization for CNC program execution.

4. **Complete G-Code Pipeline**: The exporter generates ISO 6983-compliant CNC programs from toolpath data with proper header/footer structure, coordinate formatting, and machine control commands.
5. **Interactive Design Interface**: The PyQt5-based GUI with PyVista visualization provides intuitive, real-time feedback for surface and toolpath design, enabling rapid iteration and validation.

5.1 Limitations and Future Work

The current implementation operates within several constraints that represent opportunities for enhancement:

- **Single-Axis Tool**: The Z-only toolpath generation assumes vertical tool axis (3-axis milling). Extension to 5-axis machining would enable arbitrary tool orientations and access to complex surfaces.
- **Single Tool Size**: The implementation fixes tool radius for the entire program. Support for tool-changing and adaptive tool radius would optimize machining time for varying feature geometries.
- **No Collision Detection**: The current system does not detect tool/spindle collisions with workpiece material. Implementing signed-distance field (SDF) collision checking would enable multi-pass finish strategies.
- **Limited Surface Smoothing**: While collinearity filtering reduces code length, advanced spline smoothing (e.g., cubic Hermite interpolation of toolpath points) could further improve surface finish quality and reduce machine vibration.
- **Static Machine Configuration**: Machine parameters (spindle speed, feed rates) are constant throughout execution. Adaptive feed rate control based on local surface curvature and material properties would enhance efficiency.

Future development priorities include (1) 5-axis toolpath support enabling machining of complex undercut surfaces, (2) multi-tool library with automatic tool selection based on feature geometry, (3) collision detection and clearance analysis during toolpath planning, (4) advanced path smoothing and optimization algorithms, and (5) integration with commercial CAD systems (STEP/IGES import) and CNC controllers (post-processor development for specific machine models).

The modular architecture of the codebase facilitates these extensions: new surface generation methods can be added as provider modules, and the toolpath generation pipeline supports alternative scanning strategies beyond zigzag patterns.

In summary, this project demonstrates the feasibility and practical utility of an integrated NURBS surface modelling and CNC programming system within the Python ecosystem, establishing a foundation for advanced geometric computation and manufacturing automation research.

Disclosure

This project utilized AI-assisted development tools, including Codex and Claude, during various stages of the application's design, development, debugging, and documentation. These tools provided valuable assistance in code generation, explanation, and refinement. All core design decisions, mathematical implementations, and project architecture remain the direct responsibility of the development team.

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